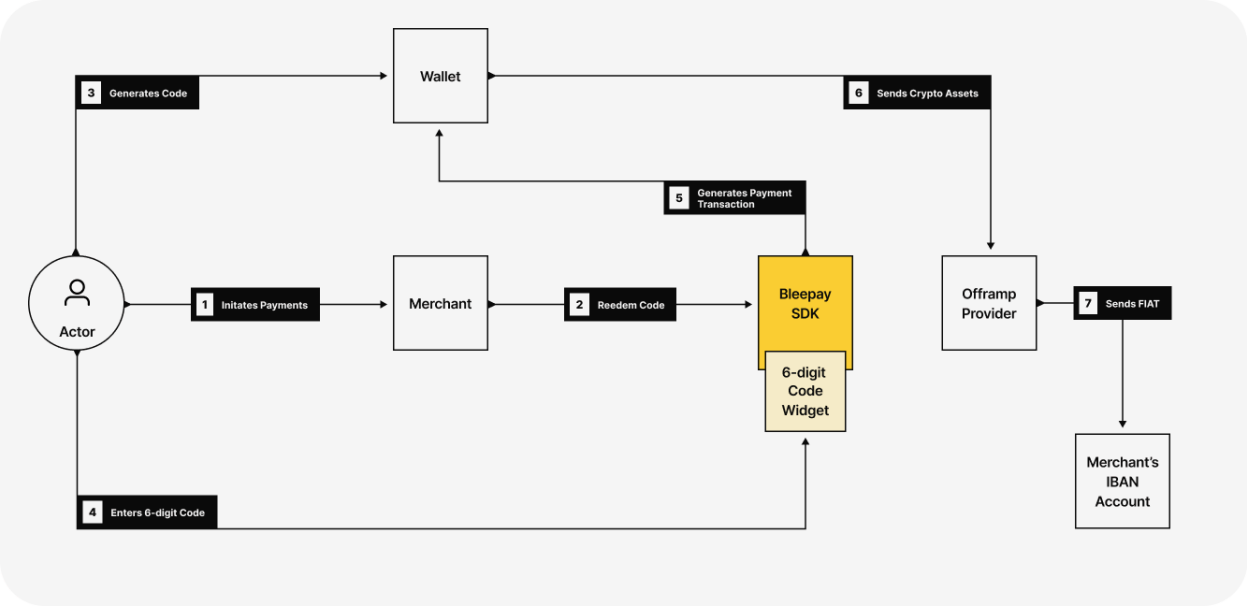
Technical Summary

1. Intro

Bleepay is a non-custodial orchestration layer that connects Web3 liquidity to traditional fiat rails. It allows users to execute payments using **any token on any chain** via human-friendly 6-digit codes, with final settlement delivered as FIAT to the merchant’s bank account through regulated partners.

Core Execution Flow

1. The merchant app triggers the Bleepay SDK to start a payment session.
2. The user generates a 6-digit session code in their non-custodial wallet, creating a secure link between their assets and the merchant's intent.
3. The merchant redeems the code via the SDK, synchronizing the transaction metadata with our backend.
4. The SDK generates the specific transaction calldata (swap/bridge/transfer) and requests the user's signature.
5. Upon signature, the wallet broadcasts assets directly to the liquidation partner's infrastructure.
6. The liquidation partner executes an automated crypto-to-fiat conversion and dispatches funds to the merchant's IBAN.



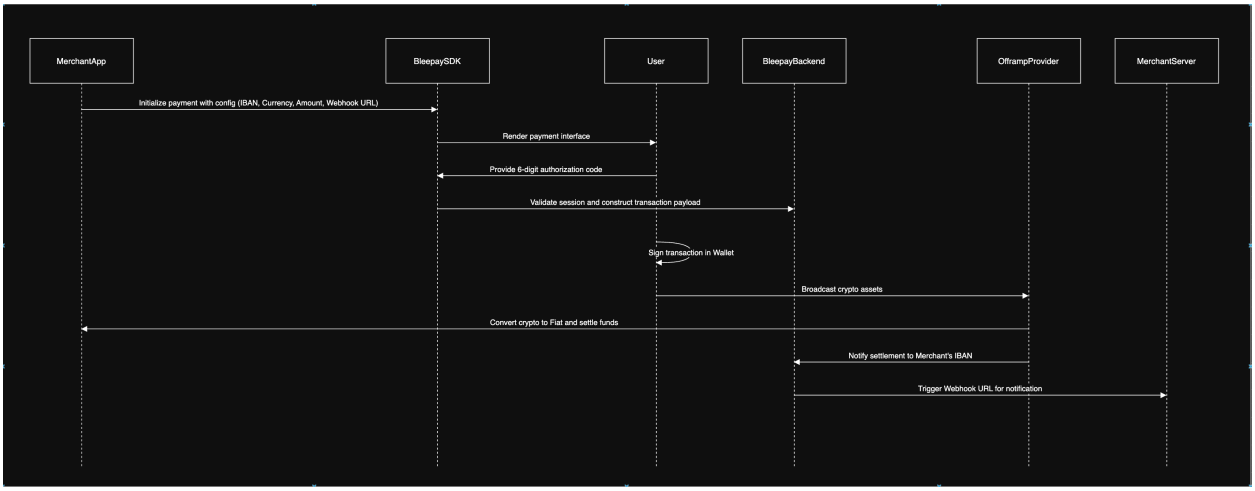
2. Integration

Bleepay is engineered to be a "drop-in" solution. We have abstracted the entire Web3 stack including gas management, cross-chain routing, and DEX interactions into a standard Web2 workflow. For the developer, Bleepay is functionally equivalent to integrating Stripe or PayPal.

Implementation Workflow

Integration is handled through standard financial metadata. There is no requirement for the merchant's engineering team to write smart contracts or manage blockchain infrastructure.

1. The merchant app initializes the SDK with standard values: amount, currency, target IBAN, and a notification webhook.
2. The SDK renders a white-labeled UI. The user enters their 6-digit code, and Bleepay handles the chain-native logic. The merchant remains entirely chain-agnostic.
3. Our backend manages the "heavy lifting"—calculating the optimal route to swap source tokens into stablecoins and bridging them to the off-ramp provider.
4. Once the bank transfer is initiated, Bleepay sends a standard JSON webhook to the merchant’s server to finalize the order in their internal system.



Example Integration Interface (Frontend SDK)

Starting a transaction requires a single method call. Developers interact only with familiar financial parameters.

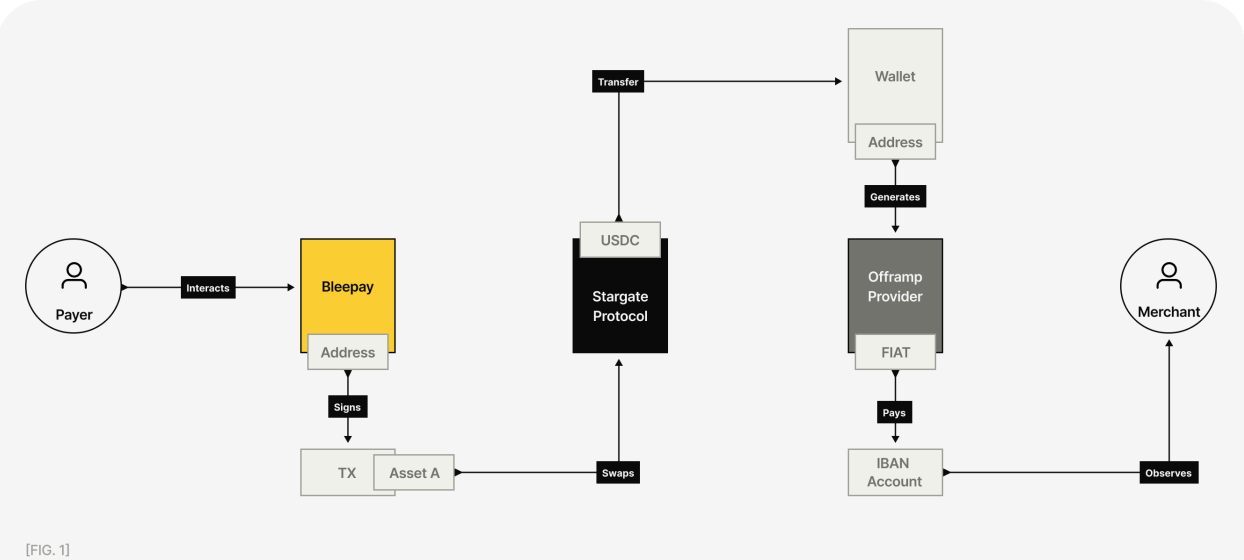
```
// Initialize the Bleepay Payment Experience
const bleepay = new BleepaySDK();

bleepay.reedem({
  amount: 299.99,              // Transaction amount
  currency: 'EUR',             // Settlement currency
  merchant_iban: 'DE12...',  // Where the Fiat funds will be
delivered
  webhook_url: 'https://api.yourshop.com/v1/payments/webhoo
k',
  order_id: 'ORDER-5678'       // Your internal reference
});
```

3. Offramp Payment Processing

Bleepay bridges the gap between digital assets and bank accounts through an intent-based processing engine. We utilize **Stargate Intent Swaps** to normalize any incoming token into **USDC/USDT** (or other compliant stablecoins) on the target liquidation chain.

Protocol eliminates "gas friction" by automatically providing necessary native tokens if a user has sufficient funds but lacks the gas required for transaction execution.



[FIG. 1]

Web2 MVP Moneyflow

The Liquidation Pipeline

- Off-ramp API generates one-time deposit addresses mapped directly to a partner's IBAN.
- Stargate executes the swap and bridge in a coordinated flow, delivering USDC directly to the liquidation point.
- The provider monitors these addresses, triggers an instant conversion to fiat upon detection, and dispatches funds via SEPA or ELIXIR.

This entire flow is monitored in real-time to manage slippage, optimize network fees, and ensure AML/KYC compliance at the liquidation gate.

4. Web3 Processing

Bleepay's architecture allows it to function as a **General Execution Layer** for any on-chain action. We are not limited to simple transfers. We can orchestrate any arbitrary smart contract interaction.

- Developers can use the Bleepay 6-digit UX to trigger complex actions like **token swaps, yield staking, NFT minting, or DAO governance**.
- We can bundle multiple interactions into a single authorized session (e.g., *"Swap ETH for USDC and immediately deposit into an Aave vault"*). Bleepay handles the complex calldata construction and presents the user with a human-readable summary instead of raw hex data.
- This removes the technical learning curve for end-users. Advanced DeFi interactions are now as simple as entering a code, with Bleepay handling gas optimization and cross-chain routing behind the scenes.

5. Dependency Management

To prevent vendor lock-in, the Bleepay architecture is strictly modular:

- **Interchangeable Off-ramps:** The fiat liquidation rail is provider-agnostic. We can swap providers based on geographic needs, fee structures, or specific merchant-signed agreements.
- **Pluggable Bridging:** While Stargate is our current default for its reliability and speed, the bridging layer is modular. We can plug in alternative protocols to ensure continuous service availability and mitigate third-party protocol risks.

6. Security

Bleepay follows a **Zero-Trust Non-Custodial Model**, strictly separating orchestration from asset custody:

- **Client-Side Signing:** Private keys remain in the user's wallet. Bleepay provides transaction payloads, but all cryptographic signatures occur locally.

- **Session Integrity:** 6-digit codes are ephemeral and bound to specific Payment Intents, preventing replay attacks or unauthorized redirection.
- **Real-time Validation:** Our engine verifies fiat requirements against on-chain swap parameters before signing to prevent price manipulation or slippage exploits.
- **Progressive Decentralization:** To eliminate single points of failure, our roadmap moves orchestration from a centralized engine to a **distributed node network**, achieving a fully trustless state.

7. Fee Structure

Transaction costs are categorized into three layers:

- **Network & Routing:** ~0.06% baseline + LayerZero/gas (Atomic swap & bridge to USDC/USDT).
- **Liquidation & Settlement:** 0.5% – 1.5% (Stable-to-fiat FX and SEPA/banking rails).

Commercial terms are finalized during onboarding with selected liquidity and settlement partners.